SER 2027 · 12TH WORLD CONFERENCE ON ECOLOGICAL RESTORATION · LISBON · LANDS IN MOTION

RABBITS, WILDCATS, POND TURTLES AND A RIVER THAT STAYED

One open connectivity screen for trophic and hydrological rewilding in the Greater Côa Valley, Portugal

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Preprint (synthesis, hypotheses, plate method)
doi.org/10.5281/zenodo.22997241
Preprint (connectivity workflow)
doi.org/10.5281/zenodo.22979020
Data: Côa Valley Eco-Connectivity v6
data.gouv.fr/datasets/coa-valley-eco-connectivity-v6
Poster code and files
github.com/coa-connectivity-lab/ser2027

THREE ANIMALS, ONE CHAIN

The European rabbit (Oryctolagus cuniculus) is the keystone prey of the Iberian Peninsula, and Endangered in its native range. The European wildcat (Felis silvestris) depends on it. The pond turtle links the pair to the river: it holds on where summer pools hold water. Rewilding projects usually treat predators, prey and rivers as three questions. In the Côa they meet in three animals.

LAND

Grazing and fire → Open mosaic → Rabbit → Wildcat

prey buffer: when rabbits crash, predators turn to reptiles (H6)

WATER

Free-flowing Côa → Summer pools → Pond turtle → River health

The chain draws on about 45 references (Angulo Lopez, 2026a) and yields seven testable hypotheses, each tied to a step in the open workflow.

METHOD

(i) Study area of 10,230 km², the Côa and 30 km around it, extended to its catchment.
(ii) Ten focal species in land, water and air groups, from GBIF records and Natura 2000.
(iii) Circuit theory in pure Python, a moving-window solver after Prima et al. (2024). It runs on a laptop, with no paid software.
(iv) Three local penalties: fire weighted by recency (MODIS, 2015 to July 2025), 22 barriers recorded in the field in August 2026, and a flat penalty over the two World Heritage areas.
(v) Rabbit screen: scrub-pasture mosaic (ESA WorldCover 2021), not burned since 2023, 200 m or more from a road, inside the top third of the land-group current, in blocks of 1 km² or more.
(vi) Pond turtle screen: pools with water in all 12 months of 2021 (JRC Global Surface Water), within 200 m of a river, reservoirs over 50 ha left out, on the better-connected half of the water-group current.

WHERE TO SURVEY FIRST

Rabbit release candidates inside the wildcat's corridors, pond turtle pools, and the two World Heritage areas, on one map. Screening maps, not release sites: every candidate needs a field survey first.

1,045 KM² rabbit candidate area, blocks of 1 km² or more

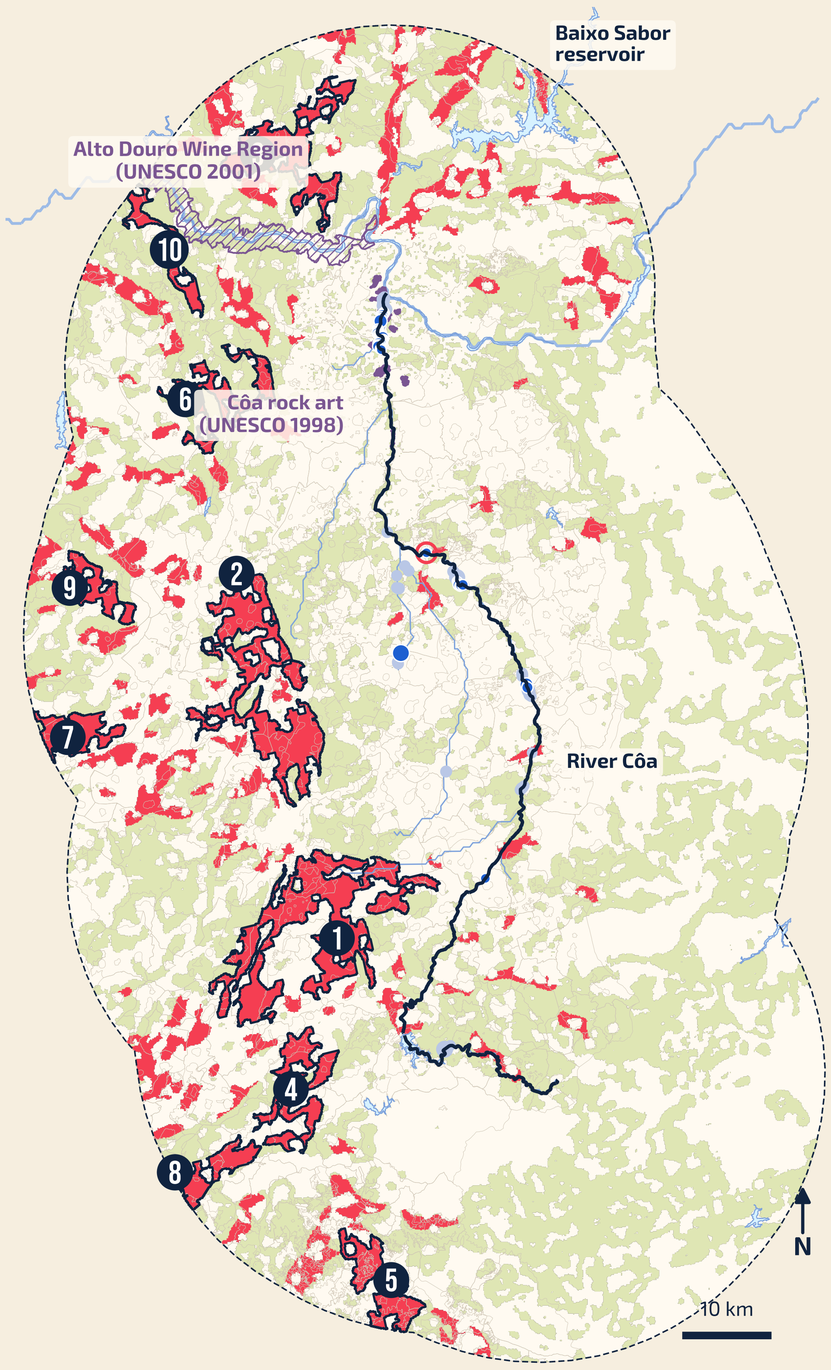
87% of it inside hunting zones

10 priority survey blocks, ranked by current

48 OF 96 year-round pools on better-connected rivers

1 pool within 2 km of a rabbit area (H6)

0 wildcat records in GBIF for the study area



Priority survey block (1 to 10)
Rabbit candidate area, survey first
Rabbit mosaic outside the corridors
Candidate pool, survey in August
Year-round pool, weaker connectivity
Pool paired with a rabbit area (H6)
Alto Douro Wine Region (UNESCO)
Côa rock-art core (UNESCO)
Reservoirs over 50 ha (left out)
Hunting zones

WHAT WE PROPOSE

(i) A rabbit baseline: repeated transects in the ten priority blocks, run with the hunters who already manage rabbits on 87% of the candidate area (H1, H3).
(ii) An August survey of the 48 candidate pools, before any pond turtle release; release only into permanent water and monitor for years (H4, H5).
(iii) For the wildcat, neutering of domestic cats near villages and mesopredator monitoring alongside any rabbit release (H2).
(iv) Add the rabbit and the pond turtle to the workflow and rerun the combined corridor (H7).

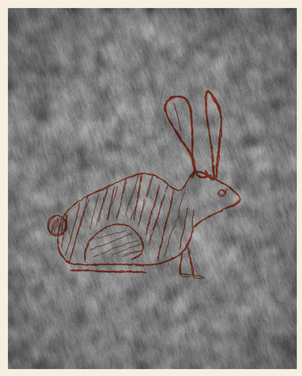
Which animals will the next chapter of the Côa include, and who decides?

WHAT THE SCREEN SHOWS

(i) Faia Brava is 88% good rabbit mosaic but carries little land-group current, so none of it makes the cut. The best corridors lie west of the river, on hunting land.
(ii) Only 1 candidate pool lies within 2 km of a candidate rabbit area. H6 can be tested at one site now; testing it widely needs a larger radius or new pools.
(iii) 13 candidate pools lie inside the Côa rock-art core. Turtle surveys there need the Archaeological Park's agreement.
(iv) 23 ha of rabbit candidate area falls inside the Alto Douro boundary. Rabbits next to vineyards is a question for growers and hunters before any release.

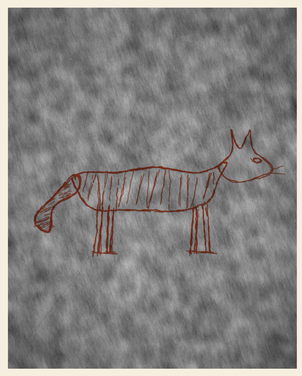
NEW ANIMALS IN AN OLD LINE

New works after the Iron Age fine-line engravings of the Côa, not ancient motifs. Each animal is reduced to one diagnostic feature and paired with one sentence of evidence. Digital studies; the hand-cut plates follow. Nothing is drawn on rock.



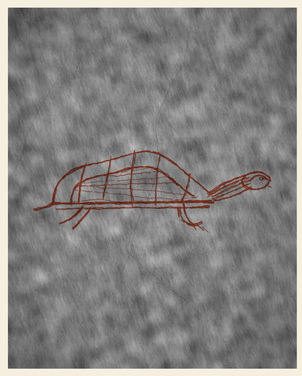
RABBIT · PREY

Keystone prey, Endangered in its native range. 1,045 km² of candidate release area. Rock art: two possible lagomorphs at Vermelhosa, Palaeolithic and uncertain.



WILDCAT · BENEFICIARY

Supported, not released: neutering of domestic cats near villages, with mesopredator monitoring. 0 GBIF records here. Rock art: three possible felids, Palaeolithic and uncertain.



POND TURTLE · RIVER INDICATOR

Presence in the Côa unconfirmed. GBIF holds 6 Emys orbicularis records (2002 to 2010) and no Mauremys leprosa. Survey in August before any release. Not in the rock art.

A RIVER THAT STAYED

The Côa stayed free-flowing because of its rock art. In 1995 students at the Vila Nova de Foz Côa secondary school took up the slogan "As gravuras não sabem nadar" (the engravings can't swim) against a dam that would have flooded open-air engravings more than 20,000 years old. The government led by António Guterres suspended the dam on 17 January 1996. UNESCO inscribed the sites in 1998, and extended them to Siega Verde, Spain, in 2010.

Where the Côa meets the Douro, the Alto Douro Wine Region (UNESCO 2001, 24,600 ha of terraced vineyard, 66 km² of it in the study area) lives on a dammed river: about 1,800 MW on the Douro mainstem by 1985, and the Baixo Sabor dam in service since 2016 on the next tributary east.

- **1995** Students take up "As gravuras não sabem nadar"
- **1996** Côa dam suspended (17 Jan); park opens
- **1998** Côa rock art inscribed by UNESCO
- **2001** Alto Douro Wine Region inscribed
- **2010** Extension to Siega Verde, Spain
- **2016** Baixo Sabor dam in service
- **2026** This screen

HYPOTHESES (Angulo Lopez, 2026a, §6). H1 Rabbits follow the grazing mosaic. H2 Wildcat habitat tracks rabbit habitat. H3 Fire resets the rabbit mosaic, for better and worse. H4 Water connectivity is limited by summer flow, not by barriers alone. H5 Pond turtles persist where summer pools persist. H6 A rabbit baseline protects the turtle. H7 The rock-art corridor is also the best combined corridor.

Angulo Lopez, L. (2026a). Rabbits, wildcats, pond turtles and a river that stayed. Preprint, Zenodo, doi:10.5281/zenodo.22997241. Angulo Lopez, L. (2026b). Movement, Water, Connection: a reproducible Python connectivity workflow. Preprint, Zenodo, doi:10.5281/zenodo.22979020. Côa Valley Eco-Connectivity v6 (CCL_v6), Etalab 2.0, data.gouv.fr. Prima, M.-C. et al. (2024). Methods in Ecology and Evolution. Bertolero, A. and Oro, D. (2009). ESA WorldCover 10 m 2021 v200 (Zanaga et al., 2022), CC BY 4.0. JRC Global Surface Water v1.4 (Pekel et al., 2016, Nature 540). HydroRIVERS v1.0 (Lehner and Grill, 2013). NASA MODIS MCD64A1. OpenStreetMap contributors (ODbL). ICNF hunting zones. DGPC classified heritage boundaries. GBIF.org occurrences, generalised to 5 km. Reis, M. (2012-2014). Inventário dos sítios da arte rupestre do Vale do Côa, Portvgalia 33-35. UNESCO World Heritage Centre, sites 866 and 1046. Resolução do Conselho de Ministros n.º 14/1996. Map: EPSG:3763; the screen is computed in scripts/poster.py from the v6 bundle. Code written with AI assistance (Claude, Anthropic) from the author's method and direction, and reviewed by the author.

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